AISHWARYA VIJAYAN

774-357-9709 • aishwaryav88@gmail.com•

EDUCATION

University of Illinois at Chicago May 2025
Doctor of Philosophy in Computer Science, 3.2 GPA

Anna University India May 2011
Master of Science in Computer Science, 3.1 GPA

Kerala University India May 2009
Bachelor of Technology in Computer Science Engineering, 3.0 GPA]

SKILLS

- Programming: C, C++, Python, Java, MATLAB
- Machine Learning: Feature engineering, modeling, evaluation, data mining
- Tools: NLP frameworks, scientific computing, graph/network analysis
- Other: Strong communication, mentoring, curriculum development

TEACHING EXPERIENCE

Roosevelt University Chicago January 2025–Present
Adjunct Faculty

- Teach graduate level and undergraduate level courses in Data Mining and Operating Systems.
- Develop instructional materials and support graduate-level learning.

University of Illinois Chicago
Teaching Assistant Fall 2015, Spring &Fall 2016-2021

- Database Systems and Advanced Database Management Systems under Prof. Ouri Wolfson.
- Programming in C/C++ under Prof. Joe Hummel
- Database systems under Prof. Aravinda Prasad Sistla

SCT College of Engineering India May 2012–June 2013
Assistant Professor (Ad Hoc)

- Taught courses in C Programming and Database Design.

MES College of Engineering India July 2011– May 2012
Assistant Professor

- Taught undergraduate/graduate courses in Database Design and Data Structures.
- Mentored students in research projects.

RESEARCH EXPERIENCE

University of Illinois Chicago

Graduate Research Assistant August 2022 – January 2023

- Conducted research on human brain data with a focus on understanding mechanisms related to consciousness.
- Performed structured data extraction from unstructured textual sources to support analytical workflows.
- Studied and implemented various prompt engineering techniques for optimizing language model performance.
- Analyzed and compared the effectiveness of prompt engineering methods across different large language models.

Graduate Research Assistant Summer 2016, 2017, 2018

- Conducted data-driven research on human brain networks using traffic-assignment models such as System Optimum and User Equilibrium
- Investigated coordination and congestion in brain networks under unconscious states using structural and functional connectomes.
- Developed Markov Decision Process (MDP)–based models to analyze dynamic signal routing in the brain

SSN College of Engineering India January 2011-March 2011
Research Assistant

- Designed IP-based WSN architecture using 6LoWPAN to support remote sensor access via HTTP without translation layers.

PUBLICATIONS

• V. Aishwarya and V.S.F. Enigo, "IP based wireless sensor networks with web interface," ICRTIT 2011.

• O. Wolfson, P. Szczurek, A. Vijayan et al., "A Traffic Analysis Perspective on Communication in the Brain," MDM 2017.

• A. Vijayan, "Deciphering Neural Codes: A Resource Search Network Perspective on Brain Connectivity," ACAI 2023.

• A. Vijayan, "A Prompt Engineering Approach for Structured Data Extraction from Unstructured Text," ACAI 2023.

REFERENCES

- Prof. Ouri Wolfson, UIC, Computer Science owolfson@gmail.com
- Prof. A. Prasad Sistla, UIC, Computer Science sistla@uic.edu
- Prof. Anastasios Sidiropoulos, UIC Computer Science sidiropo@gmail.com
- Prof. Olusola Ajilore, Psychiatry, University of Illinois Chicago oajilore@uic.edu
- Prof. Alex Leow, Psychiatry & Bioengineering UIC alexfeuillet@uic.edu
- Dr. Jerry Schnepp, Roosevelt University, jschnepp@roosevelt.edu

Dear Members of the Grossman Center Search Committee,

I am writing to apply for the Postdoctoral Fellow position in Theoretical and Computational Neuroscience at the Grossman Center for Quantitative Biology and Human Behavior at the University of Chicago. I earned my Ph.D. in Computer Science from the University of Illinois Chicago, where my research combined computational modeling, network science, optimization and data-driven analysis to understand information flow in large-scale brain networks using structural (DTI) and functional (fMRI) connectivity data. I am excited about the Grossman Center's emphasis on mathematical and computational approaches to understanding neural circuits and cognition and about the opportunity to work in UChicago's interdisciplinary environment.

My research has focused on coordination and decision-making in complex networks. In my doctoral work, I analyzed how brain activity patterns align with global versus local optimization principles by adapting traffic-assignment ideas (System Optimum vs. User Equilibrium) to neural communication constrained by connectome structure. I also studied dynamical and Markov Decision Process (MDP) inspired routing models to study how signals propagate under uncertainty and how coordination changes across brain states. This training has prepared me to build mechanistic models grounded in data, derive testable predictions and iterate between theory and empirical evaluation.

At the Grossman Center, I am especially interested in pursuing the following three project directions, each aligned with specific Grossman Center laboratories:

1. **ML + modeling with experimental collaborators (Muscinelli Lab)**
   I am interested in projects that combine machine learning with mechanistic modeling to explain neural data. I would like to work on building predictive models that connect brain activity to underlying circuit structure and constraints and to develop hybrid pipelines that use both data-driven ML and theory-based models in collaboration with experimental datasets.
2. **How biological constraints shape computation (Noorman Lab)**
   I would like to study how biological constraints such as connectivity and resource limits shape brain dynamics and cognitive function. I am interested in modeling how circuits perform efficient and flexible computation under real constraints and how these constraints limit or enable behaviors such as navigation and planning.
3. **Representations and abstraction in noisy networks (Nogueira Lab)**
   I am also interested in understanding how neural systems form stable representations and abstractions despite noise. I would like to work on projects that analyze the geometry of representations in neural data and in trained models and to study how abstraction emerges as networks learn under constraints.

Methodologically, I enjoy projects that combine theory and modeling with careful evaluation deriving predictions from models, testing them against neural and behavioral data and refining models to improve explanatory power. I am comfortable working with large-scale network data, building simulation pipelines and collaborating across disciplinary boundaries.

Thank you for considering my application. I would welcome the opportunity to discuss how my background in network modeling, optimization and data-driven neuroscience can contribute to the Grossman Center's research community.

Sincerely,
Aishwarya Vijayan, Ph.D.
aishwaryav88@gmail.com | 774-357-9709
https://scholar.google.com/citations?user=ozWPKeIAAAAJ&hl=en